Startup Guide

FANUC CR-35*i*A Series

Thank you for purchasing a FANUC Robot.

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

- Straight-head screwdriver
- Cross-head screwdriver
- Torque wrench
- Metric hex-head wrenches

Required Items

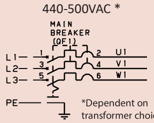
- Properly rated transport equipment as specified
- Customer-supplied properly rated power source as specified on the Controller Power Source Label
- Customer-supplied properly rated input power cable; conductor size of AWG10; terminal size is M5
- Multimeter
- 12-M16 x 65mm bolts
- 12-M16 washers
- Rigid platform to hold robot
- Loctite #243

Optional Items

- Strain relief cable grip
- Robot locator pins

Three Phase

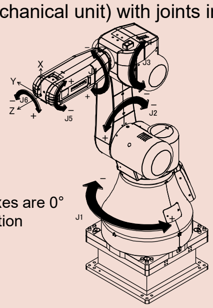
440-500VAC *



*Dependent on transformer choice.

Controller

Robot (mechanical unit) with joints indicated



CR-35*i*A shown

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A. Unpack/Install your Controller


Your shipment will arrive on two skids. One will contain the Robot Mechanical Unit. The other contains the Controller, and Accessory Boxes.

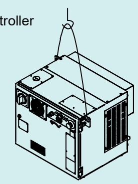
Shipment Contents

Hardware Bag, Sensor Calibration CD, Teach Pendant, & Inspection Data Sheet

1. Remove the bolts holding the top of the container and remove the top. Carefully loosen the bolts on each side and remove one side at a time. Unwrap and properly dispose of the plastic around the robot. Open all of the boxes included in the shipment.

2. Use properly rated equipment to remove the controller from the skid.

Minimum crane capacity: 300kg
Minimum sling capacity: 300kg



> ⚠ **WARNING**
> **Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.**

3. Place the controller in a safe location where the grounding wire, RMP cable, and sensor cable can reach the robot, and the controller is safely reachable from outside of the robot work area.

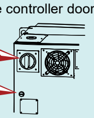
1 - Sensor cable (square connector)
2 - Teach Pendant cable (round connector)
3 - RMP cable (rectangular connector)
NOTE: this cable is also called the Robot Connection Cable (RCC)
4 - Ground wire

4. Connect the controller to power:

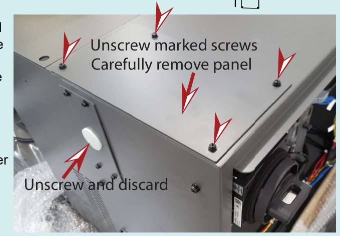
> ⚠ **WARNING**
> **Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.**

a. Determine the power requirements for your controller from the label on the front. The label is at the bottom left corner of the controller door.

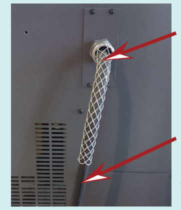
b. Open the controller door by using a straight-head screwdriver to turn the latch clockwise, then turn the Breaker counter-clockwise past its off position.



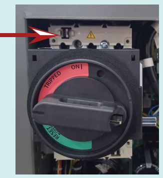
c. Unscrew the round white plug from the side of the controller. Remove the four screws highlighted in the illustration to the right, and carefully remove the Breaker access panel. It may stick as it has a sealant around the edges.



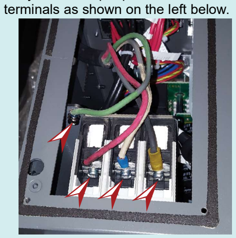
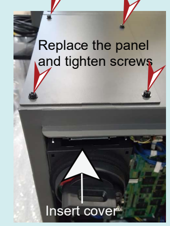
d. Route the user-supplied input power cable through the side hole. It is typical to purchase and install a strain relief cable grip (example shown below) to remove stress on the power cable connections and isolate the power cable from the controller cabinet.



e. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.



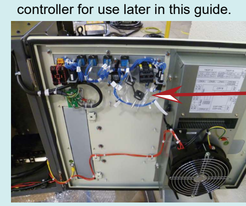
f. Verify that the input power cable is not energized. Connect it to the terminals as shown on the left below.

g. Re-insert the Breaker cover and click it into place. Place the Breaker access panel back over top of the breaker and secure with four screws.

h. With the breaker still off, energize the input power cable and use a multimeter to verify the correct voltage is present via the holes in the breaker cover.

i. Before you close the controller door, cut loose the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.

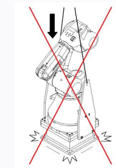
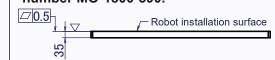


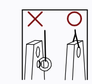
j. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller counter-clockwise.

Controller installation is complete.
DO NOT TURN ON THE CONTROLLER UNTIL YOU ARE INSTRUCTED TO DO SO. Go to **B. Unpack and Install your Robot.**

B. Unpack and Install your Robot

> ⚠ **WARNING**
>
> **Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.**
>
> **When hoisting or lowering the robot with a crane or forklift, move it slowly with great care. When placing the robot on the floor, exercise caution to prevent the installation surface of the robot from striking the floor forcefully.**
>
> **It is extremely important that the installation plate is flat after securing to the floor. Ensure there is no gapping between the installation plate and the floor; also confirm no debris is between the mounting surfaces. Fill any open spaces or gaps, especially directly underneath the robot base, with grout or similar method, so that the plate does not flex or become warped and so no gaps are present after installation. The installation surface should not be painted or fabricated. The installation plate is FANUC part number MO-1800-590.**
>
> **Always use special transport brackets and attach sling properly.**

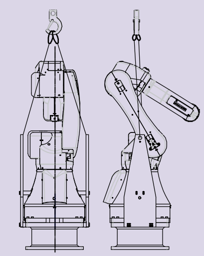





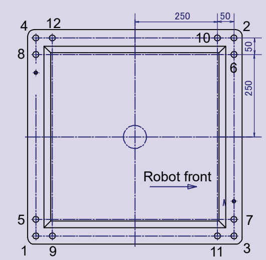
1. Remove the bolts holding the robot to the skid.

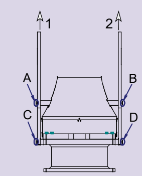
2. Use a properly rated crane or forklift and included special transportation brackets to hoist the robot to its new location.

Crane capacity min: 2.5 ton
Sling capacity min: 1.0 ton/sling
Sling length: 3+ meters



3. Thread 12-M16 washers and 12-M16 bolts (tensile strength 1200N/mm² or more) through the robot base into the base plate, but leave them a little loose. Loosen the transport bracket bolts marked A and B below. Using the crane, support the first bracket, remove the bolts marked C, and remove the bracket. Support the second bracket with the crane, remove bolts marked D, and remove the bracket. Tighten the robot mounting bolts in the numeric order indicated below to a torque spec of 319Nm. Store the transport brackets and special skid for future transportation. More information is available in the *CR-35iA Mechanical Unit Operator's Manual.*





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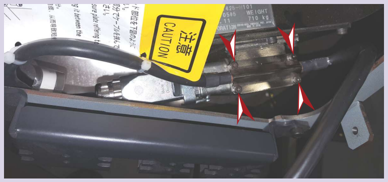
Startup Guide

FANUC CR-35*iA* Series

4. Remove the four hex bolts from the back cover, and remove the cover.

5. Connect the force sensor cable as shown below. Remove the four hex bolts and pressure plate. Secure the exposed shield of the sensor cable under the pressure plate and tighten the hex bolts.



6. Connect the RMP cable and ground wire as shown to the right. These cables are also referred to as the Robot Connection Cables (RCC). Secure all the cables with zip ties.

7. Reattach the back cover with the four hex bolts removed in step 4.

8. Connect the teach pendant to its cable as shown below. Rotate the teach pendant cable until it is seated. Then, spin the end of the connector until it is threaded on tight. Wiggling the connector might let you tighten it more.

Robot installation is complete. You are now ready to go to **C. Turn on your Controller**.

C. Turn on your Controller

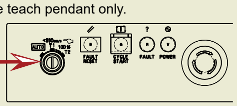
⚠ WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

1. Turn on the controller by rotating the Breaker clockwise to ON.

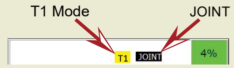
2. Locate the Mode switch on the Controller Operator Panel. Use the key retrieved in section **A** step **4i** to switch to T1 mode. T1 mode limits the robot speed to 250 mm/sec and requires robot control to be at the teach pendant only.

Mode Switch



3. Verify the controller is in T1 Mode by looking at the top right corner of the teach pendant screen. The mode indicator will be highlighted yellow and look similar to the status bar below. If it is not T1, repeat the previous step.

4. Press the COORD key on the teach pendant until JOINT is displayed similar to the status bar to the right.



Teach Pendant Status Bar

You have turned on your controller. You are now ready to go to **D. Jog your Robot**.

D. Jog your Robot

NOTE: To continue setup, the controller needs to have been **powered on for 30 minutes** or more. After this warmup time, continue below.

1. Collaborative robot settings

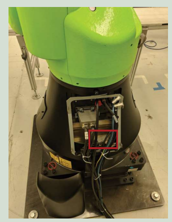
a. Using a computer, load the .cm file from the Sensor Calibration CD (see Step 1) to a USB disk.

b. Insert the USB into the robot controller and navigate to where the .cm file was saved. (MENU>>FILE>>File. F5, [UTIL]>>Set Device>>USB Disk (UD1:)).

c. With the .cm file highlighted, press ENTER to execute the .cm file. You will be asked if you want to execute the file, press F4, Yes.

d. Once the file is loaded, the sensor serial number should be shown in the DCS Collaborative robot screen. This number should match the collaborative sensor serial number that is physically on the robot.

```
DCS
   Collaborative robot                  1/28
                                      Status
   Contact stop status:          SAFE
   Enable/Disable: ENABLE              OK
   Force sensor
     Serial number:        161110096   OK
   Group:                          1   OK
   Payload setup:           <DETAIL>   OK
   Active Payload: No. 1 [ZERO           ]
   External force Limit  / Disabling input
    @Limit 1:    70.00[N]  ---[  0]    OK
```



e. Turn the teach pendant ON/OFF switch to the ON position. Press MENU, cursor to NEXT, and press ENTER. Cursor to SYSTEM > Motion and press ENTER. On No. 1, press F3, DETAIL.

f. Cursor to Payload [kg], press 0, then press ENTER. You will prompted to answer the question shown below. Press F4, YES. Optionally, you can cursor up one and type a name such as "ZERO" for identification.

```
MOTION/PAYLOAD SET
                                        2/8
   Group 1
 1 Schedule No[    1] [ZERO              ]
 2 PAYLOAD              [kg]          0.00
 3 PAYLOAD CENTER X     [cm]          0.00
 4 PAYLOAD CENTER Y     [cm]          0.00
 5 PAYLOAD CENTER Z     [cm]          0.00
 6 PAYLOAD INERTIA X    [kgfcms^2]    0.00
 7 PAYLOAD INERTIA Y    [kgfcms^2]    0.00
 8 PAYLOAD INERTIA Z    [kgfcms^2]    0.00

 Path and Cycletime will change.  Set it?
                                  YES    NO
```

g. Press the PREV key and press F5, SETIND. Type 1 for the payload number, and press ENTER. This makes schedule 1 the active payload.

h. Press MENU, choose NEXT, cursor to SYSTEM > DCS, and press ENTER. Press F2, APPLY, you will be prompted for the DCS code number. Enter the default of 1111.

```
DCS
                                       1/18
  1 Collaborative robot:               CHGD
  2 Safe I/O Status:
  3 Safe I/O connect:                  OK
  4 Joint position check:     ----     OK
  5 Joint speed check:        ----     OK
  6 Cart. position check:     UNSF     OK
  7 Cart. speed check:        SAFE     OK
  8 T1 mode speed check:               OK
  9 User model:                        OK
 10 Tool frame:                        OK
 11 User frame:                        OK
 Code number(master): ----
 [ TYPE ]   APPLY   DETAIL        UNDO
```

i. On the following screen, press F4, OK, to continue.

```
--- Collaborative robot ---------------
 Enable/Disable: ENABLE            OK
 Force sensor
   Serial number:      150810511   OK
                         ALL   OK   QUIT
```

j. You must cycle power on the controller. Turn the Breaker off, wait 5 seconds, and then turn it back on. Alternatively, you can press the FCTN key, cursor to 0 NEXT, and press ENTER. Then cursor to 8 CYCLE POWER, press ENTER, and select YES.

k. After the robot has cycled power, the screen to the right will be displayed. Press F2, CONFIRM, you will be prompted for the DCS code number. Enter the default of 1111.

```
DCS
   Collaborative robot                  1/22
                                      Status
   Contact stop status:          DSBL
   Enable/Disable: ENABLE              OK
   Force sensor
     Serial number:        150810511   OK
   Group:                          1   OK
   Payload setup:           <DETAIL>   CHGD
   Active Payload: No. 1 [ZERO           ]
   External force Limit  / Disabling input
    @Limit 1:   150.00[N]  ---[  0]    OK
     Limit 2:     0.00[N]  ---[  0]    OK
 Code number(master): ----
 [ TYPE ]  CONFIRM  PEAK CLR  [CHOICE]  UNDO
```

l. You will be prompted with the current payload schedule. If No. 1 is displayed similar to below, confirm by pressing F4, YES. If another number is displayed press F5, NO, and repeat the procedures starting at section **D** step **1e**.

```
Actual payload is No.1:ZERO?
                              YES    NO
```

m. You will be prompted to check that no person or object is in contact with the robot arm. When verified, press F4, YES.

```
Nobody contacts with the robot?
                              YES    NO
```

You should see "Payload confirmation success" at the bottom of the screen. Continue to the next step.

2. General settings

a. Press MENU, select 0 (Next), and select (SYSTEM), and then select CONFIG.

- Use the teach pendant arrow keys to move the cursor to the Hand Broken option, and press ENTER.
- Use the teach pendant function keys to disable the hand broken setting for each motion group. **NOTE:** It might already be disabled on your system.

b. Press MENU, select 0 (Next), select (SYSTEM), and then select CONFIG.

- Use the teach pendant arrow keys to move the cursor to the Enable UI signals option.
- Use the teach pendant function keys to set the option to FALSE.

c. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.

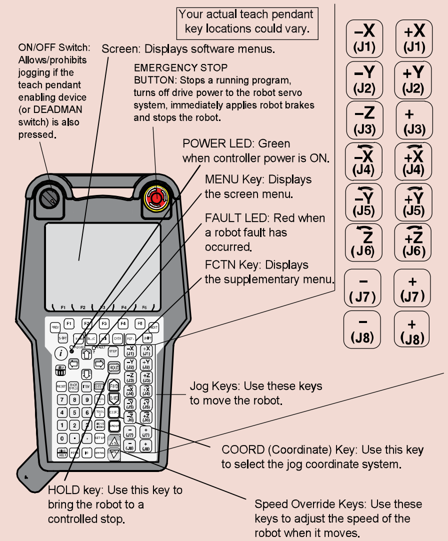
3. Press and hold a DEADMAN switch on the back of the teach pendant to its middle position. Press RESET on the teach pendant to clear alarms.

NOTE: To view alarms, press MENU, cursor to ALARM, and press ENTER. To toggle between alarm history and active alarms, press F3.

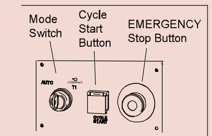
To move each joint axis of the robot, press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the **Robot (mechanical unit) with joints indicated** diagram on Page 1 of this guide.

Try jogging each joint of the robot.

Turning on and Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the *Controller-specific Software Installation* manual.

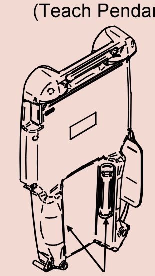


Refer to your *Application-Specific Setup and Operations Manual* for more information on each teach pendant key.



Controller Operator Panel

*i*Pendant (Teach Pendant)



Teach Pendant Enabling Device (also called DEADMAN switch).

For more information

To access controller, robot, or application-specific manuals, scan the **QR Code** to the right or go to **https://www.fanucamerica.com/**.

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